

# Curriculum Vitae (CV)

Seyed Saber Naserlavi

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## Personal Information

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## Education

- Diploma: Allameh Helli High School, Tehran, Mathematics, 2000
- Bachelor's degree: Shahid Bahonar University of Kerman, Civil Engineering, 2004
- Master's degree: Iran University of Science and Technology, Civil Engineering, Highway and Transportation Engineering, 2006.
- PhD: Tarbiat Modares University, Civil Engineering, Highway and Transportation Engineering, 2011

## Skills

### Programming Languages & Scientific Computings

- **Python:** (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, SciPy)
- **R & RStudio:** (dplyr, ggplot2, caret, Shiny, tidyverse, tidymodels, mlr3)
- **Julia**
- **MATLAB**

## Machine Learning & Artificial Intelligence

- **Core ML/DL Frameworks:** TensorFlow, Keras, PyTorch, Scikit-learn
- **Techniques & Models:**
  - Deep Learning,
  - Supervised & Unsupervised Learning (Regression, Classification, Clustering, Dimensionality Reduction)
  - Reinforcement Learning (e.g., for Adaptive Traffic Signal Control)
  - Natural Language Processing (NLP) (e.g., Text Classification, Sentiment Analysis, Topic Modeling)
  - Computer Vision (e.g., Vehicle and Pedestrian Detection for Safety Analysis)
  - Explainable AI (XAI) & Model Interpretability Techniques
- **ML Platforms & Tools:** IBM SPSS Modeler, Neural Designer, RapidMiner Studio

## Data Analysis, Statistics & Visualization

- **Statistical Software:** IBM SPSS Statistics, IBM SPSS AMOS, EViews, GraphPad Prism, Design-Expert, LISREL, GAMS, Lingo
- **Data Visualization:** Tableau, SigmaPlot, (Python libraries like Matplotlib/Seaborn, R's ggplot2)
- **Advanced MS Excel:** (Data Analysis Toolpak, Solver, Power Query, VBA scripting for data manipulation)

## Software Engineering & Research Tools

- **Version Control:** Git, GitHub/GitLab.
- **Documentation & Typesetting:** LaTeX, EndNote, Mendeley, MathType
- **Project Management:** MS Project
- **Cloud Computing Concepts & Distributed Processing**

## Operating Systems & General Productivity

- **Operating Systems:** Windows, macOS, Linux
- **Microsoft Office Suite:** Word, PowerPoint, Access, MS Excel (*Advanced MS Excel capabilities detailed under Data Analysis, Statistics & Visualization*)

## Methodologies & Advanced Techniques

- **Metaheuristic Optimization for solving transportation network design problems:** Proficient in designing, implementing, and applying a range of metaheuristic algorithms (e.g., Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Grey Wolf Optimizer (GWO), Simulated Annealing) for solving complex, multi-objective optimization problems.
- **Systematic Review & Meta-Analysis:** Expertise in conducting comprehensive systematic literature reviews and performing quantitative meta-analyses (e.g., using CMA) to synthesize research findings and identify evidence-based insights.
- **Structural Equation Modeling (SEM):** Proficient in developing, estimating, and interpreting complex theoretical models using SEM techniques (e.g., Confirmatory Factor Analysis (CFA), Path Analysis, Latent Growth Curve Modeling) with software such as AMOS and LISREL.
- **Advanced Statistical Modeling & Inference:**
  - Generalized Linear Models (GLMs): Including logistic regression, Poisson regression, and other extensions of linear models for various data types.
  - Multivariate Statistical Analysis: (e.g., MANOVA, Factor Analysis, Cluster Analysis).
  - Time Series Analysis & Forecasting: (e.g., ARIMA, GARCH models).
  - Survival Analysis / Reliability Analysis: (e.g., Kaplan-Meier, Cox Proportional Hazards).
- **Fuzzy Logic Systems & Soft Computing:** Design and application of fuzzy set theory, fuzzy inference systems, and hybrid fuzzy models for handling uncertainty, linguistic variables, and imprecise reasoning in complex systems.
- **Resilience & Sustainability Modeling:** Developing and applying quantitative and qualitative frameworks and models to assess, predict, and enhance the resilience and sustainability of complex systems (e.g., infrastructure, socio-ecological systems).
- **Discrete Choice Modeling:** Expertise in designing stated and revealed preference surveys, and estimating various discrete choice models (e.g., Multinomial Logit (MNL), Nested Logit, Mixed Logit) to understand and predict decision-making behavior.
- **Queuing Theory & Simulation Modeling:**
  - Application of queuing theory principles for performance analysis of service systems.
  - Development of Discrete-Event Simulation (DES), Agent-Based Modeling (ABM), or System Dynamics (SD) models for “what-if” scenario analysis, system optimization, and bottleneck identification (e.g., using VISSIM, AIMSUN, or custom scripts).
- **Transportation Network Analysis (Graph Theory):** Utilizing network theory and algorithms to model, analyze, and visualize complex relationships, system structures, and information flow.

- **Experimental Design & Data Collection:** Proficient in designing robust experimental and quasi-experimental studies, survey instrument development (e.g., for choice experiments), and implementing rigorous data collection protocols.

## Interest

- study, research, learning something, academic activities, industry and agriculture.
- Sports, entertainment, games, and PlayStation.

## Awards

- The first place in the under-12 chess tournament of Kerman province
- The first place in the under-14 chess tournament in Kerman province and participating in the national tournament in Babol
- The first place in the junior table tennis tournament of Kerman province
- Futsal champion of Shahr Bahonar University of Kerman (attending as coach, main player, and team captain)
- Passing the school entrance exam of the National Organization for the Development of Talents (on average, one out of every 1500 students was accepted.)
- The first place in the entry exam for the doctoral course in the field of civil engineering, highway and transportation, 2007.
- The first rank of the graduates of the doctoral course of the Faculty of civil and Environmental Engineering, Tarbiat Modares University, 2011.
- Received a certificate of appreciation from the command of the police force of Kerman province, research week, 2010.
- Received a certificate of appreciation in the first national conference on traffic safety and implementation strategies for its improvement, Kerman, 2010.
- Received a certificate of appreciation and a special research awarded statue for the best paper in the research topic of highway geometric design in the 16th International Conference on Transportation and Traffic Engineering of Iran, Tehran, 2016.
- Received a certificate of appreciation from the Artificial Intelligence Development Headquarters; Vice Presidency for Science and Knowledge-Based Economy, designing and presenting a course titled “Applied Artificial Intelligence in Traffic and Transportation Engineering”, the selected project of the national call for the compilation of artificial intelligence educational syllabi in various academic disciplines, Tehran, September 2024.

## Research Area

1. traffic safety modeling

- severity / frequency accident modeling
  - surrogate safety measure
2. Traffic flow modeling / traffic engineering
  3. Transportation / travel behavior modeling
  4. Geometric design of highways
  5. Machine learning / Data mining application
  6. pavement engineering

## Publications

### Refereed Journal Publications

1. **Seyed Saber Naseralavi**, Shirgir Behrooz, Kazemi Tehrani Hamid, New technique of micro-simulation of traffic flow: Concepts of Cellular Automata in traffic flow, Scientific Quarterly of Traffic Engineering Promotion of Tehran Transport and Traffic Organization, number 38, pages 56-61, 2009.
2. **Seyed Saber Naseralavi**, Saffarzadeh Mahmoud, Mamdohi Amirreza, Nadimi Navid, Development of Urgent Declaration Safety Indicator Designed for Opportune Detection of Rear-End Conflict, Scientific-Research Quarterly of Transportation Engineering, Year 1, Number 3, Pages 27-39, Spring 2009. ([link](#))
3. **Seyed Saber Naseralavi**, Saffarzadeh Mahmoud, Adresi Mostafa, Comparison of road safety evaluation methods based on traffic crash information and traffic conflicts, Scientific and Promotional Rahvar Quarterly, Volume: 6, Number: 6, pages 23-40, summer 2009. ([link](#))
4. **Seyed Saber Naseralavi**, Saffarzadeh Mahmoud, Airline Schedule Planning Using Heuristic Search Method, scientific-research quarterly of Transportation Research Journal, -6th year, 2nd issue (serial number: 19), summer 2009, p. 175-184. ([link](#))
5. Mamdohi Amirreza, Shirgir Behrouz, Ebadi Zeinab, **Seyed Saber Naseralavi**, Investigation of the effect of ramp metering on traffic flow in a highway section using simulation: a case study of a section of Aye Elah Sadr highway, Traffic Engineering Journal, Tehran Transport and Traffic Organization, number 40, pages 50-55, 1388.
6. Saffarzadeh Mahmoud, **Seyed Saber Naseralavi**, Applying Time-to-Collision in Collision Avoidance Systems, Journal of Traffic Engineering, No. 43, pages 10-16, 2010. ([link](#))
7. **Seyed Saber Naseralavi**, Saffarzadeh Mahmoud, Nadimi Navid, Estimation of Car-following Behavior using Fundamental of Differential Equations, Journal of Transportation Research, Volume:7, Issue: 3, Pages 261 to 274, 2011. ([link](#))
8. Taleghani Seyed Ali, Shahbaznia Mehdi, **Seyed Saber Naseralavi**, Identification and Prioritization of Delay Factors in Road Construction Projects in Iran, Journal of Traffic Engineering, Volume:11, Issue: 43, pages 17-28, 2010. ([link](#))

9. Hadigeh Javani Mohsen, Saffarzadeh Mahmoud, **Seyed Saber Naseralavi**, The Study of a Prioritization Model to Increase the Safety of Pedestrian Walkways Using Artificial Neural Networks, Traffic Management Studies, Volume: 5, Issue: 16, pages 39-54, 2010. ([link](#))
10. **Seyed Saber Naseralavi**, Saffarzadeh Mahmoud, Nadimi Navid, Mamdohi Amirreza, Applying Time-to-Collision to Enhance the Coefficient of Determination for GHR Car-Following Model in Deceleration Mode, Omran Modares scientific-research journal, 11th volume, number 1, pages 17-26, 2011. ([link](#))
11. **Seyed Saber Naseralavi**, Saffarzadeh Mahmoud, Mamdohi Amirreza, Nadimi Navid, Simulation and its application in traffic studies; car chase model, Rahor Quarterly, 8th year, serial number 15, pp. 59-73, 1390
12. **Seyed Saber Naseralavi**, Nadimi Navid, How to Use Traffic Safety Indicators for Freeway Safety Evaluations?, Journal of Traffic Engineering, Volume: 12, Issue: 46, pages 18-24, 2012. ([link](#))
13. **Seyed Saber Naseralavi**, Noughabi, M Gh., Ayati, E., Modeling Multiple-Vehicle Property Damage Collisions in Urban Signalized Intersections, International Journal of Transportation Engineering (IJTE), Vol. 1, No. 1, pp. 35-46, Winter 2012. ([link](#))
14. Saffarzadeh, M., Nadimi N., **Seyed Saber Naseralavi**, Mamdohi A. R. A General Formulation for Time-To-Collision Safety Indicator, proceedings of the institution of civil engineers (ICE)-transport (ISI), Volume 166, Issue TR5, pp 294-304, 2013. ([link](#))
15. **Seyed Saber Naseralavi**, Saffarzadeh Mahmoud, Mamdohi Amirreza, Nadimi Navid, Traffic simulation in urban health-safety, Bimonthly Artificial Intelligence and Precision Tools, Year 6, Number 6 (36 consecutive), pp. 6-2, 2012. ([link](#))
16. Nadimi, Navid, **Seyed Saber Naseralavi**, Bagheri, Ramtin, Headway Analysis for Different Combination of Vehicles in a Congested Traffic Flow in freeways, Journal of Transportation Engineering, Volume: 3, Issue: 4, Pages 379-386, 2013. ([link](#))
17. **Seyed Saber Naseralavi**, Nazari Hamed, Izadi Amir, Evaluation of the effect of separation in light and heavy vehicles on safety indices by car-following situation, Journal of Traffic Engineering, Volume:13, Issue: 53, 2013. ([link](#))
18. Jafari Nasab Seyed Ehsan, Pourmalem Nasser, Seyed Saber Naseralavi, Quantitative Safety Assessment of Vehicular Ad-hoc Networks (VANETs) in Freeway Traffic, Traffic Management Studies, Volume: 9, Issue: 33, Pages 1-21, 2014. ([link](#))
19. **Seyed Saber Naseralavi**, Saffarzadeh Mahmoud, Nadimi Navid, Comparison of Time to Collision and Time Headway in Rear End Collision Avoidance Systems, Traffic Management Studies, Volume: 9, Issue: 34, Pages 1-18, 2015. ([link](#))
20. S. Ehsan Jafari Nasab, **Seyed Saber Naseralavi**, Nasser Pourmoallem, Investigating the Influence of Vehicular Ad-hoc Networks on Driver Behaviors and Traffic Accidents, Travel Behavior and Society Volume 2, Issue 3, Pages 156-165, September 2015. ([link](#))
21. Behbahani, H., Nadimi N., **Seyed Saber Naseralavi**, New time-based surrogate safety measure to assess crash risk in car-following scenarios, Transportation Letters: the International Journal of Transportation Research, VOL 7, NO 4, Pages 229-238, 2015. ([link](#))
22. Shojaei, BM, Ismaila, A., **Seyed Saber Naseralavi**, Firoozi, AA, Performance evaluation

- of Highway base stabilized with styrene - butadiene copolymer latex and Portland cement, International Journal of Pavement Research and Technology, Volume 9, Issue 4, July 2016, Pages 321-336. ([link](#))
23. Shojaei, BM, Ismaila, A., **Seyed Saber Naseralavi**, Firoozi, AA, Freeze-thaw performance of base course treated with carboxylated styrene-butadiene emulsion-Portland cement, International Journal of Pavement Engineering, 2016, Pages 1-9. ([link](#))
  24. Mohsen Poor Arab Moghadam, Parham Pahlavani, **Seyed Saber Naseralavi**, Prediction of Car Following Behavior Based on the Instantaneous Reaction Time using an ANFIS-CART Based Model, International Journal of Transportation Engineering (IJTE), Vol.4, No.2, Pages 109-126, Autumn 2016. ([link](#))
  25. **Seyed Saber Naseralavi**, Nadimi Navid, Nakhai, Mohsen, Development of Macroscopic Queuing Model for Evaluating Effect of Capacity Changing on Vehicle Delays, Journal of Transportation Engineering Research, Volume: 11, Issue: 3, Serial Number 44, Pages 719-735, April 2020. ([link](#))
  26. Mahmoud Saffarzadeh, Akram Mazaheri, **Seyed Saber Naseralavi**, Development of a new integrated surrogate safety measure for applying in intelligent vehicle systems, International Journal of Transportation Engineering (IJTE), Volume:8 Issue: 4, 2021, PP 383-398. 10.22119/IJTE.2021.260315.1547. ([link](#))
  27. Nadimi Navid, Asadamraji Morteza, **Seyed Saber Naseralavi**, Calculating dynamic thresholds for critical time to collision as a safety measure, Proceedings of the Institution of Civil Engineers – Volume 175 Issue 7, December, 2022, pp. 403-412. ([link](#))
  28. Ali Reza Ghanizadeh, **Seyed Saber Naseralavi**, An Explicit Formulation for Estimation of Structural Number (SN) of Flexible Pavements in 1993 AASHTO Design Guide using Response Surface Methodology (RSM), Journal of Soft Computing in Civil Engineering, Volume 6, Issue 2 - Serial Number 20, April 2022, Pages 35-50. ([link](#))
  29. Abdi Behnam, **Seyed Saber Naseralavi**, Validity and reliability of drivers anger scale and its relationship with psychological parameters and evaluation of differences based on age and sex (Case study: Tehran metropolis). Scientific-Research Journal of Modares Civil Engineering, Volume 22, Number 6, Pages 75-86, 2021. ([link](#))
  30. Mazaheri Akram, Saffarzadeh Mahmoud, **Seyed Saber Naseralavi**, Nagrawi Ahmad, Determining the best location of traffic indicators with the aim of estimating travel time with genetic algorithm, Scientific Journal of Traffic Engineering, serial number 85, page 61-77, 2022.
  31. Mohammad Moein Brahui Jahanshahi, **Seyed Saber Naseralavi**, Investigation of the effect of complying with the speed limit in reducing traffic conflicts in urban sections, Scientific Journal of Traffic Engineering, serial number 87, page 1-8, 2022.
  32. Mazaheri Akram, Saffarzadeh Mahmoud, **Seyed Saber Naseralavi**, Development of Surrogate Safety Measure Based on the Concept of Vehicle Influence Area, Journal of Transportation Engineering Research, Volume: 14, Issue: 2, Serial Number 55, Pages 2355-2364, January 2023. ([link](#))
  33. Nadimi Navid, **Seyed Saber Naseralavi**, Amirhossein Zare Mirhosseini, Evaluation of applying Fuzzy Inference System to increase the efficiency of intelligent vehicles by Surrogate Safety Measures, Scientific-Research Quarterly of Transportation Journal,

- Volume: 19, Issue: 71, Serial Number 71, Pages 31-46, January 2022. ([link](#))
34. Mohammad Soltani Rad, **Seyed Saber Naserlavi**, Akram Mazheri, Modeling the severity of pedestrian-vehicle accidents using multinomial logit regression, Traffic Management Studies, Volume:17, Issue: 66, page 1-34, 2023. ([link](#))
  35. **Seyed Saber Naserlavi**, Mohammad Soltani Rad, Akram Mazheri, Modeling the severity of motor-vehicle accidents by different age groups of the driver using multinomial logit regression. Transportation Engineering Research Quarterly, Articles in Press, Accepted Manuscript, Available Online from 31 October 2022. ([link](#))
  36. Akram Mazaheri, Saffarzadeh, M., Nadimi N., **Seyed Saber Naserlavi**, A Revise on using Surrogate Safety Measures for Rear-End Crashes, journal of International Association of Traffic and Safety Sciences (IATSS Research), Volume 47, Issue 1, April 2023, Pages 105-120. ([link](#))
  37. Ali Reza Ghanizadeh, **Seyed Saber Naserlavi**, Intelligent Prediction of Unconfined Compressive Strength and Young's Modulus of Lean Clay Stabilized with Iron Ore Mine Tailings and Hydrated Lime Using Gaussian Process Regression, Journal of Soft Computing in Civil Engineering, Volume 7, Issue 4 - Number 26, October 2023, Pages 1-23. doi:10.22115/SCCE.2023.370814.1573. ([link](#))
  38. **Seyed Saber Naserlavi**, Navid Nadimi, Seyed Amir Mohammad Hosseini, Modeling Railway Crash Severity Using Multinomial Logit Method Based on Freight and Passenger Trains, Journal of Transportation Research, Volume:21 Issue: 2, Pages 59-76, 2024. ([link](#))
  39. Ali Reza Ghanizadeh, F Safi Jahanshahi, **Seyed Saber Naserlavi**, Intelligent modelling of unconfined compressive strength of cement stabilised iron ore tailings: a case study of Golgohar mine, European Journal of Environmental and Civil Engineering, 1-29, 2023. ([link](#))
  40. Ali Reza Ghanizadeh, Farzad Safi Jahanshahi, **Seyed Saber Naserlavi**, Application of Gene Expression Programming for Modeling Bearing Capacity of Aggregate Pier Reinforced Clay, International Journal of Mining and Geo-Engineering, Volume 58, Issue 1, Pages 113-119, March 2024.
  41. **Seyed Saber Naserlavi**, Navid Nadimi, Seyed Amir Mohammad Hosseini, Modeling the severity level of truck crashes using binary logistic regression method, Submitted to Research Quarterly Transportation Research Journal, 2023.
  42. Navid Nadimi, **Seyed Saber Naserlavi**, Vahid Khalife, Mehdi Aminian, Analysis and evaluation of the factors of road users' injuries using the extraction of associational rules, Research Quarterly Journal of Transportation Research, published online since 2024 May 26, 10.22034/TRI.2024.430
  43. Sadegh Muslimi Mohni, Ali Nadran, **Seyed Saber Naserlavi**, Comparison of machine learning algorithms in feature selection and severity classification of pedestrian accidents (Case study: Qom city), Research Journal of Transportation Research Quarterly, published online since 2024 June 16.
  44. Navid Nadimi, Alireza Zare, Vahid Khalifeh, **Seyed Saber Naserlavi**, Identifying safety issues of vulnerable road users in urban streets and optimizing countermeasures through the integration of association rule mining and mathematical programming methods, Research Journal of Transportation Research Quarterly, Accepted Manuscript Available

Online from 13 January 2025. ([link](#))

45. **Seyed Saber Naserlavi**, M. Soltanirad, M. Igene, K. Jimenez, E. Ranjbar, (2025). Determining Factors Affecting Motorcycle Crash Severity Based on Time of Day and Season Combinations, *ENG Transactions* 6, 1–11, Article ID: 2941.
46. **Seyed Saber Naserlavi**, Mohammad Soltanirad., Erfan Ranjbar, Keshav Jimenez, Martin Lucero, Mahdi Baghersad and Akram Mazaheri. (2025). A Short-Term Storytelling Framework for Understanding Surrogate Safety Measures in Intelligent Vehicle Interactions. *Future Transp.*, 5, 86. <https://doi.org/10.3390/futuretransp5030086>.
47. **Seyed Saber Naserlavi**, M. Soltanirad, M. Baghersad, E. Ranjbar, K. Jimenez, A. Mazaheri, (2025). Exploring the Impact of Driver Sex, Driver Age, Area Type, and Lighting Conditions on Rear-End Collision Severity, *Computational Research Progress in Applied Science & Engineering*, CRPASE: Transactions of Civil and Environmental Engineering 11, 1–16, Article ID: 2957.
48. **Seyed Saber Naserlavi**, Soheil Rezashoar, Akram Mazaheri, Saman Shafaati, ”Enhancing Short-Term Traffic Flow Forecasting by Hybrid Deep Learning Architectures and Attention Mechanisms: A Case Study on the High-Density Karaj-Chalous Road, Iran”, *International Journal of Transportation Engineering (IJTE)*, Articles in Press, Accepted Manuscript, Available Online from 27 September 2025, 10.22119/ijte.2025.500027.1683.
49. **Seyed Saber Naserlavi**, Mohammad Soltanirad, Erfan Ranjbar, Martin Lucero, Fateme Gorzin, Yasaman Hakiminejad, Shiva Azimi, Mahdi Baghersad, Akram Mazaheri, “Machine Learning Modeling of Household Trip Generation by State Using NHTS Data”, *Urban Science*. 2025; 9(9):353. <https://doi.org/10.3390/urbansci9090353>.
50. **Seyed Saber Naserlavi**, Mohammad Soltanirad., Erfan Ranjbar, Martin Lucero, Mahdi Baghersad, Mehran Piri, Mohammad Javad Hassan Zada, and Akram Mazaheri. (2025). “Modeling the Severity of Crashes in Rainy Weather by Driver Gender and Crash Type” *Future Transportation* 5, no. 4: 146. <https://doi.org/10.3390/futuretransp5040146>.

## Conference Papers

1. Saffarzadeh, M., **Seyed Saber Naserlavi**, Modern Roundabouts as an Alternatives to conventional Intersections, Presented at 7th Transportation & Traffic National Conference, Tehran, 2007. ([link](#))
2. Saffarzadeh, M., **Seyed Saber Naserlavi**, Comparison of HCM & FDOT Methodology regarding Operation Performance of Arterial Roads, Presented at 7th transportation & traffic National conference, Tehran, 2007. ([link](#))
3. Saffarzadeh, M., **Seyed Saber Naserlavi**, Hasaninasab, S., Roundabout Delay Estimation with Simulation Models: DRACULA, VISSIM, and SITRA-B+, Presented at Third International Symposium of Ttransport Simulation 2008. ([link](#))
4. Saffarzadeh, M., **Seyed Saber Naserlavi**, Hasaninasab, S., Eliminating left turns at signalized intersections using adjacent roundabouts, Presented at 8th Transportation & Traffic National Conference, Tehran, 2009. ([link](#))

5. Mamdoohi, A.R., **Seyed Saber Naserlavi**, Urban Transportation Management from the Two Dimensions of Supply & Demand: Case of Telecommunicating, Presented at 1st Innovative Ideas in Urban Management Conference, Tehran, 2009.
6. Saffarzadeh, M., **Seyed Saber Naserlavi**, Hasaninasab, S., Evaluation of New Urban Intersection Traffic Control Designs, 1st Innovative Ideas in Urban Management Conference, Tehran, 2009.
7. **Seyed Saber Naserlavi**, Shirgir, B., Alipour, S., Mohseni, H., Evaluation & Comparison of Lane Changing Models in Microscopic Traffic Simulation, 9th Transportation & Traffic National Conference, Tehran, 2010. ([link](#))
8. Saffarzadeh, M., **Seyed Saber Naserlavi**, Nadimi, N., Future of Traffic Control & Management: Introduction to Vehicular Ad Hoc Networks, 9th Transportation & Traffic National Conference, Tehran, 2010. ([link](#))
9. Mamdoohi, A.R., Shirgir, B., Ebadi, Z., **Seyed Saber Naserlavi**, Analysis of Ramp Metering Impact on Traffic Flow using Simulation: Sadr Highway Case Study, 9th Transportation & Traffic National Conference, Tehran, 2010. ([link](#))
10. Kavussi, A., Adressi, M., **Seyed Saber Naserlavi**, Evaluation of Roughness Indicators for Airport Pavement, 9th international conference of civil engineering, Shiraz, 2010. ([link](#))
11. **Seyed Saber Naserlavi**, Saffarzadeh, M., Mamdoohi, A.R., Evaluation & Validation of Traffic Flow Simulation, 2nd Internal Safety & Traffic Conference, Tehran, 2010.
12. **Seyed Saber Naserlavi**, Saffarzadeh, M., Nadimi, N., The Position of Traffic Simulation in Urban Health: Traffic Safety, The 10th International Conference on Traffic and Transportation Engineering, Tehran, 2010. ([link](#))
13. Ghasemi, N. M., Jalili G. M., **Seyed Saber Naserlavi**, Ahmadi, M., Shaghaghi, F. N., Prioritizing the Factors affecting the rate of fatigue and sleepiness of heavy vehicles drivers using TOPSIS Method, The 1st National Conference on Traffic Safety, Kerman, 2011. ([link](#))
14. Ghasemi, N. M., **Seyed Saber Naserlavi**, Ghasemi, N. M., Statistic report of the fatigue and sleepiness accidents of vehicles drivers using police report in scene accident, The 1st National Conference on Traffic Safety, Kerman, 2011. ([link](#))
15. **Seyed Saber Naserlavi**, Baghersad, M., Naghdi, M., Nadimi, N., Conflict based Traffic Safety Studies: Literature Reveiw, The 1st National Conference on Traffic Safety, Kerman, 2011. ([link](#))
16. **Seyed Saber Naserlavi**, Baghersad, M., Taleghani, A., Micro-simulation based Traffic Safety Studies: Literature Reveiw, The 1st National Conference on Traffic Safety, Kerman, 2011. ([link](#))
17. **Seyed Saber Naserlavi**, Hassanpour, S., Nadimi, N., Adressi, M., study on TTC application in safety related analysis, The 1st National Conference on Traffic Safety, Kerman, 2011. ([link](#))
18. **Seyed Saber Naserlavi**, Nadimi N., A Generalized Formulation for Time-To-Collision Safety Indicator and its Application, 11th International Conference on Transportation & Traffic, Tehran, 2011. ([link](#))
19. Behbahani, H., Nadimi, N., **Seyed Saber Naserlavi**, Traffic Simulation and its

- Application in Traffic Studies: Car-Following Models, The 11th International Conference on Traffic and Transportation Engineering, Tehran, 2011. ([link](#))
20. Behbahani, H., Nadimi, N., Amini, A.A., **Seyed Saber Naserlavi**, Determination of driver instantaneous reaction time using GHR car following model, The 11th International Conference on Traffic and Transportation Engineering, Tehran, 2011.
  21. Ghasemi, N. M., Jalili G. M., Ahmadi, M., **Seyed Saber Naserlavi**, Sadeghi, A., Operational implementation program for decreasing fatigue traffic crashes, 9th International Congress of Civil Engineering, Isfahan, 2012. ([link](#))
  22. **Seyed Saber Naserlavi**, Saffarzadeh, M., Malmir, H., Time headway variations just in lane changing time, The 12th International Conference on Traffic and Transportation Engineering, Tehran, 2012. ([link](#))
  23. **Seyed Saber Naserlavi**, Nazari, H., Ghaderi, Sh. M., Development of a model for determination of number of rural drivers based on social and driver behavioural characteristics, 1st national conference on evaluation of rural accidents, Kerman, 2012.
  24. **Seyed Saber Naserlavi**, Mahmoudi M., M., Kargar D., J., Development of a model for optimum off-road parking locationing using AHP, The 13th International Conference on Traffic and Transportation Engineering, Tehran, 2013.
  25. **Seyed Saber Naserlavi**, Forozan, M., Economical analysis of highway rest areas using ARENA simulation software, The 13th International Conference on Traffic and Transportation Engineering, Tehran, 2013. ([link](#))
  26. **Seyed Saber Naserlavi**, Nazari, H., Izadi, A., Titidezh, O., Accident frequency prediction of rural roads based on social and driver behavioural characteristics, The 13th International Conference on Traffic and Transportation Engineering, Tehran, 2013. ([link](#))
  27. Malmir, H., **Seyed Saber Naserlavi**, Saffarzadeh, M., Time headway analysis of different type of pairing vehicles by lanes, national conference of modern computational techniques and optimization methods in civil engineering, Saghez, 2013. ([link](#))
  28. Malmir, H., **Seyed Saber Naserlavi**, Saffarzadeh, M., analysis and modeling of headway of different type of pairing vehicles in congested flow of freeway, national conference of modern computational techniques and optimization methods in civil engineering, Saghez, 2013. ([link](#))
  29. Seyed Ehsan Jafari Nasab, Nasser Pourmoallem, **Seyed Saber Naserlavi** Reduction of Freeway Accidents through Using of Intelligent Vehicular Networks, International conference on civil engineering, architecture and Urban Sustainable Development, 2014. ([link](#))
  30. Seyed Ehsan Jafari Nasab, **Seyed Saber Naserlavi**, Exploring the VANETs Vehicular Ad hoc Networks Impact on Traffic Conflicts, The 14th International Conference on Traffic and Transportation Engineering, Tehran, 2014.
  31. **Seyed Saber Naserlavi**, Nazari, H., Izadi, A., Titidezh, O., Effect of vehicle type on surrogate safety measure in the car following situation, The 14th International Conference on Traffic and Transportation Engineering, Tehran, 2014.
  32. **Seyed Saber Naserlavi**, Evaluation of design methods of concrete pavements for highways and airports, 1st national conference on concrete pavements, TMU, Tehran, 2015.

33. **Seyed Saber Naserlavi**, Evaluation of modern construction methods of concrete pavements, 1st national conference on concrete pavements, TMU, Tehran, 2015.
34. Nadimi, N., **Seyed Saber Naserlavi**, Amraji, M., Sayadi, M., Determination of challenges and research needs in rural transportation and presentig some strategies for implementing results, 1st national conference on rural transportation, Tehran, 2015.
35. **Seyed Saber Naserlavi**, Nadimi, N., Evaluation of environmental effects of rural roads, 1st national conference on rural transportation, Tehran, 2015.
36. **Seyed Saber Naserlavi**, Nadimi, N., Evaluation of freeway safety management system and, 4th international conference on highway accidents, Tehran, 2015.
37. **Seyed Saber Naserlavi**, Nadimi, N., Shafiei, M., Comparison of car following models, The 15th International Conference on Traffic and Transportation Engineering, Tehran, 2015.
38. **Seyed Saber Naserlavi**, Izadi, A., Nazari, H., Sarabi, M., Evaluation surrogate safety index and selecting the most appropriate index for In Vehicle Warning System, The 15th International Conference on Traffic and Transportation Engineering, Tehran, 2015.
39. Mohsen Poor Arab Moghadam, Parham Pahlavani, **Seyed Saber Naserlavi**, Applying Adaptive Neuro Fuzzy Inference system and Regression Tree for Prediction Car Following Behavior Based on Instantaneous Reaction Time, The 15th International Conference on Traffic and Transportation Engineering, Tehran, 2015.
40. Iftikharzade Seyed Hamed, **Seyed Saber Naserlavi**, The role of intelligent transportation management systems in supporting logistics and the automotive chain, the second international conference on intelligent transportation systems in Tehran, 2015.
41. **Seyed Saber Naserlavi**, Shakiba Mohammad, Functionality improvement of driverless cars using traffic data and artificial neural networks, 2nd International Conference on Intelligent Transportation Systems, Tehran, 1395.
42. Iftikharzadeh Seyed Hamed, **Seyed Saber Naserlavi**, The need to improve productivity in the country's road freight transport by modifying fleet performance management methods, 16th International Conference on Transportation and Traffic Engineering of Iran, Tehran, 2015.
43. Ahmad Nagrawi, **Seyed Saber Naserlavi**, Investigation and comparison of response drive and safe distance car chase models, 16th International Conference on Transportation and Traffic Engineering of Iran, Tehran, 2015.
44. Mozbanzadeh Dezfouli Abdul Majid, **Seyed Saber Naserlavi**, Choosing the optimal type of non-level intersection based on traffic studies with AHP method, 16th International Conference on Transportation and Traffic Engineering of Iran, Tehran, 2015.
45. Pourarab Moghadam Mohsen, Pahlavani Parham, **Seyed Saber Naserlavi**, car-following modeling considering momentary reaction time: support vector regression methodology, 16th International Conference on Transportation and Traffic Engineering of Iran, Tehran, 2015.
46. Ahmad Nagrawi, Sharif Tehrani Saleh, **Seyed Saber Naserlavi**, Determining the optimal cycling route using network planning (studying the route from Beyhaghi terminal to Amir Kabir University of Technology), 2nd International Conference on Transportation and Traffic Engineering of Iran, Tehran, 2015.

47. **Seyed Saber Naserlavi**, Navid Nadimi, Fariborz Mansourifar, Presenting the new concept for clearance distance variation of vehicles with respect to time, 17th International Conference on Transportation and Traffic Engineering, Tehran, 2016.
48. **Seyed Saber Naserlavi**, Behnam Abdi, Using traffic conflict analysis to evaluate safety according to the conflict severity, 17th International Conference on Transportation and Traffic Engineering, Tehran, 2016.
49. Shahram Shafiei, Navid Nadimi, **Seyed Saber Naserlavi**, sandstorm cost determination modeling using the fuzzy inference system (case study of Sistan and Baluchistan Province), Second International Conference on Civil Engineering, Architecture and Crisis Management, 2016.
50. Ali Farnoud, **Seyed Saber Naserlavi**, Evaluation of the relative influence of structural and environmental factors on the performance and response of pavements with long-term pavement performance (LTPP) data, the first national road and transportation engineering conference, 2016.
51. Seyed Kamaluddin Mirjafarian, **Seyed Saber Naserlavi**, Navid Nadimi, Comparison of the probability of accidents of vehicles with different speeds on the freeway with surrogate safety indicators, 2nd International Conference on Civil Engineering, Architecture and Crisis Management, 2016.
52. Fatemeh Shahidzadeh Asadi, **Seyed Saber Naserlavi**, improvements prioritization of highway section using multi-criteria decision-making and safety audit method, the first national road and transportation engineering conference, 2016.
53. Mohammad Amin Zarei, **Seyed Saber Naserlavi**, safety evaluation model and determination of accident prone sections using accidents history (Case study: Shiraz-Saadat Shahr Center), First National Road and Transportation Engineering Conference, 2016.
54. **Seyed Saber Naserlavi**, Abdul Hamid Atut, Investigation of traffic flow modeling at the pico level, the first national conference in recent years in transportation engineering and planning, 2016.
55. **Seyed Saber Naserlavi**, Mohsen Nakhai, Investigation of station design factors in the high-speed railway network, the first national conference on transportation engineering and planning, 2016.
56. Pouria Shabani, **Seyed Saber Naserlavi**, Increasing the resilience of vital arteries by providing an optimal routing model and geometric road design, the first national conference on resilience of vital arteries and civil infrastructures, Building and Housing Research Center, 2016. ([link](#))
57. Mazaheri Akram, Saffarzadeh Mahmoud, **Seyed Saber Naserlavi**, Nagrawi Ahmad, Determining the optimum traffic detector location for travel time estimation with genetic algorithm, 18th International Conference on Transportation and Traffic Engineering, 2019.
58. Habib Taghipour, **Seyed Saber Naserlavi**, Navid Nadimi, Evaluation and ranking of factors on the creation of a smart city from transportation point of view: Shiraz city case study, the second scientific and research conference on urban planning, civil engineering, architecture and environment, France, December 20, 2019.
59. Sharifi Mohammad Sadegh, **Seyed Saber Naserlavi**, Mazaheri Akram, injury severity

- modeling of single passenger motorcycle accident using ordered logistic regression model, National Conference of Civil Engineering, Smart Development, 18 Shahrivar, 2020.
60. Aminian Mehdi, Nadimi Naveed, **Seyed Saber Naserlavi**, Evaluating the role of autonomous vehicles to improve transportation sustainability, the first national conference on civil engineering, smart development, 18 Shahrivar, 2020.
  61. **Seyed Saber Naserlavi**, Mazaheri Akram, Nadimi Navid, Investigation of naturalistic observational methods in traffic psychology research, 5th National Conference of Life Science Psychology, March 19, 2021.
  62. **Seyed Saber Naserlavi**, Alireza Ghanizadeh, Mohammad Araf Yarahamdi, a comparison of different machine learning methods for predicting the flow number of asphalt mixtures, 9th National Conference on New Materials and Structures in Civil Engineering, Noshirvani University of Technol, 2021.
  63. **Seyed Saber Naserlavi**, Seyed Amir Mohammad Hosseini, injury severity accident modeling in foggy weather conditions using random forest, 19th International Conference on Transport and Traffic Engineering, March 2021.
  64. **Seyed Saber Naserlavi**, Amir Hossein Sotoudeh, Akram Mazaheri, Evaluation of surrogate safety indicators based on principal component analysis and clustering, 19th International Conference on Transport and Traffic Engineering, March 2021.
  65. **Seyed Saber Naserlavi**, Akram Mazaheri, Evaluation of the quality ranking of universities in academic systems: an analysis of global trends, the 17th Conference on Evaluation and Quality Assurance in Academic Systems, 2024.
  66. **Seyed Saber Naserlavi**, Akram Mazaheri, Navid Nadimi, Descriptive analysis of the relationship between the severity of accidents and environmental factors using statistical analysis and hypothesis testing, 6th International Traffic Safety and Management Conference, 2024.
  67. **Seyed Saber Naserlavi**, Akram Mazaheri, Challenges of safety analysis of self-driving cars using traffic conflict technique, 6th international traffic safety and management conference, 2024.
  68. Hossein Mahmoodabadi, **Seyed Saber Naserlavi**, Akram Mazaheri, Navid Nadimi, Application of intelligent transportation systems in sustainability concept, the fourth national conference on civil engineering, intelligent development and sustainable systems, Golestan University, 2024.
  69. Zahra Arabpour, **Seyed Saber Naserlavi**, Akram Mazaheri, Navid Nadimi, Application of Machine Learning in Intelligent Transportation Systems (ITS), the fourth national conference on civil engineering, intelligent development and sustainable systems, Golestan University, 2024.
  70. Saman Shafaati, **Seyed Saber Naserlavi**, Akram Mazaheri, Reza Jalal Kamali, Analysis and development of intelligent transportation systems based on the Internet of Things: challenges and opportunities, Golestan University, 2024.
  71. Sahar Latifi, **Seyed Saber Naserlavi**, Akram Mazaheri, Reza Jalal Kamali, Big Data Analytics in Intelligent Transportation Systems: Challenges and Opportunities, Fourth National Conference on Civil Engineering, Intelligent Development and Sustainable Systems, Golestan University, 2024.

72. **Seyed Saber Naseralavi**, Akram Mazaheri, Navid Nadimi, Analysis of urban data and its impact on the environment: challenges and solutions, the first national conference on urban environment, University of Tehran, 2024.
73. Saman Shafaati, **Seyed Saber Naseralavi**, Reza Jalalkamali, Design and Narrative Techniques for Urban Data Visualization, the first national conference on urban environment, University of Tehran, 2024.
74. Saman Shafaati, Seyed Saber Naseralavi, Navid Nadimi, The Role of Bicycle Transportation in the Transition to Sustainable Cities: Challenges and Solutions for Urban Infrastructure Development, Second National Conference on Innovation in Green Industries, University of Mazandaran, 2024.
75. Soltanirad, M., **Seyed Saber Naseralavi**, Jimée, K., Luo, Q., Bataineh, T., Igene, M., Liu, H., (2025). Development of a New Safety Indicator for Predictive Safety Analysis Based on Different Arbitrary Surrogate Safety Measures Using LiDAR Sensor Data. Paper presented at the Transportation Research Board Annual Meeting, Washington, D.C. Available from TRB.
76. **Seyed Saber Naseralavi**, Akram Mazaheri, Analysis of Transportation Needs in the Economic Route of the Silk Road: An Opportunity for Sustainable Development, International Conference on the Silk Road; Cultural Roads, Shared Heritage; Kerman, 2025.
77. **Seyed Saber Naseralavi**, Akram Mazaheri, Saman Shafaati, Soheil Reza Shoar, Determining the factors affecting the severity of heavy truck accidents in California by vehicle model and AADT; 7th International Conference on Traffic Management and Safety; Tehran, 2025.
78. **Seyed Saber Naseralavi**, Akram Mazaheri, Soheil Reza Shoar, The Failure of Imported Algorithms in Iranian Traffic: Transitioning to Contextual AI, The 20th International Conference on Traffic and Transportation Engineering, Tehran, 2026.
79. **Seyed Saber Naseralavi**, Akram Mazaheri, Modern Urban Resilience in Iran: Combating Seismic and Environmental Threats via Smart Bio-Based Models, The 20th International Conference on Traffic and Transportation Engineering, Tehran, 2026.
80. **Seyed Saber Naseralavi**, Akram Mazaheri, Traffic Behavior Digital Twins (TBDT): A New Architecture for Human-Centric Transportation in Iran, The 20th International Conference on Traffic and Transportation Engineering, Tehran, 2026.

## Work Experience

1. Traffic expert, Traffic and Transportation Deputy of Kerman Municipality, 5 months.
2. Consultant of Tehran Transport and Traffic Organization, Planning and Modeling Deputy, 1 year.
3. Faculty member of the first traffic and safety conference and its implementation solutions, Kerman, April 31 and May 1, 2018.

4. Member of the faculty of the 3rd National Conference on Earthquakes and Structures, Kerman, October 26 and 27, 2011.
5. Member of the faculty of the 12th International Conference on Transport and Traffic of Iran, Tehran, 2018.
6. Member of the faculty of the 13th International Conference on Transportation and Traffic of Iran, Tehran, 2012.
7. Member of the faculty of the 14th International Conference on Transportation and Traffic of Iran, Tehran, 2013.
8. Member of the faculty of the Babol National Civil Engineering Congress, Tehran, 2012.
9. Member of the faculty of the first national congress of concrete procedures, Tehran, 2014.
10. Academic faculty member of the 19th International Conference on Transportation and Traffic, Iran, Tehran, March 1401.
11. The director of the transport safety department of Parse Designers Transport Research Institute, from 2008 to 2009.
12. Referee of many prestigious scientific journals (with more than 100 refereed articles), including:
  - IJTE (International Journal of Transportation Engineering,
  - Scientific Research Journal of Transportation Engineering,
  - Scientific Research Journal of Civil Engineering of Modares,
  - Scientific Journal of Traffic Engineering
13. Manager and responsible for various research projects related to transportation topics, including:
  - Analysis of Highway safety in Kerman province and implementation solutions to reduce accidents
  - Economic evaluation and technical analysis of welfare service complexes in suburban Highways of Kerman province
  - Providing a budget allocation model for the maintenance of the country's Highways
  - Compilation of the country's Highway safety strategic plan
  - Studies of bicycle routing in Hamedan city
  - The plan for monitoring the statistics of even and odd plan areas and simulating the traffic plan area
14. Faculty member of Civil engineering Department, Shahid Bahonar University of Kerman, Kerman, Iran.

## **Teaching and Workshops**

### **Undergraduate**

- Transportation Engineering

- Traffic Engineering
- Analysis and Modeling of Road Safety
- Data Mining and Machine Learning Techniques in Civil Engineering
- Highway Geometric design
- Pavement Engineering
- Design, Operation and Maintenance of Civil Engineering Systems
- Highway Project
- Construction Machinery
- Surveying

## **Graduate**

- Advanced Traffic Engineering
- Advanced Highway Geometric Design
- Operation Research
- Transportation Planning
- Advanced Pavement Analysis and Design
- Econometrics

## **Workshops**

- Asphalt technology
- Comparative Traffic Management Models
- Traffic Flow Simulation
- Highway Safety Evaluation
- Advanced Pavement Analysis and Design
- Machine learning using R language meta-packages of tidyverse & tidymodels
- Data Mining Application in transportation engineering
- Intelligent Transportation and Information Technology
- Principles of Reproducible research with RStudio and Quarto
- The importance of Python and R bilingualism in research

## **Book Translation**

The following books have been translated into Persian

- Transportation Infrastructure Engineering: A Multimodal Integration, by Lester A. Hoel, Nicholas J. Garber, Sadek, 2009
- Traffic Flow Fundamentals by May, 1990.
- Pavement Design and Materials, by A. T. Papagiannakis, E. A. Masad, 2008.

- Observational Before-After Studies in Road Safety: Estimating the Effect of Highway and Traffic Engineering Measures on Road Safety by E. Hauer, 1997.

## Alumni

1. Presentation of safety performance function model in urban signalized intersections, case study: Kerman city intersections, **Mehdi Baghersad**
2. Presentation of the prioritization model for Highway safety improvement activities, case study: Kerman province, **Iman Delavari**
3. time headway analysis by different combinations of vehicle pairs in conditions of dense freeway traffic flow, **Hamed Malmir**
4. Safety analysis of Highways in Kerman province and implementation solutions to reduce accidents, **Amin Karimdadi Shamsabad**
5. Evaluation and development of freeways safety indicators for Car-following Conditions based on micro-simulation traffic flow, **Hamed Nazari**
6. Prioritizing traffic areas for the off-road parking using the AHP method, **Mitra Mahmoudi**
7. Dynamic modeling of rod axial length change using shape functions with high continuity order under high frequency considering linear and non-linear behavior, **Mohammad Hassan Eshaghi, 1395/12/15 (advisor)**
8. Evaluation of the safety impact of vehicular ad hoc networks (VANETs) on freeways, **Seyed Ehsan Jafarinasab**
9. Comparison of methods for identifying accident-prone points: accident analysis (AA), road safety audit (RSA) and traffic conflict technique (TCT), **Fatemeh Shahidzadeh Asadi, 1396**
10. Development of a car-following model using freeway dense traffic flow data, **Hamid Chaichizadeh**
11. Economic evaluation and technical analysis of rest areas in suburban Highways of Kerman province, **Mohammad Forozan**
12. Presenting the life cycle cost analysis model for Highway construction management at the project level, **Ehsan Dehghanfard**
13. **Khosrow Tafrishi**
14. Identification and prioritization of accident points on Etbat Aliaat Highway in Kermanshah province and development of identification methods using multi-criteria decision making method, **Ali Amini**
15. Bearing capacity of circular foundations in layered soils, **Vahid Moulai Kohbanani, 10/11/2015. (advisor)**
16. Explosion compact effect on the liquefaction talent reduction, **Javad Mahmoudi, 1395/06/31 (advisor)**
17. Elasto Static analysis based on NURBS & Infinite Continue Functions, **Majid Mohammadi Nia, 1395/12/25**

18. Assessing and comparing the risk of accidents of heavy vehicles and passenger cars in saturated freeway traffic using microscopic traffic flow data, **Ahmad Nagrawi, 2015.**
19. Using surrogate safety measure to evaluate road safety in an uninterrupted flow facilities and developing a new one, **Mahmoudreza Nafer. (advisor)**
20. Evaluating the technical and economic performance of using interlayer asphalt to control reflective cracks, **Mohammad Moshtari**
21. Selection of non-level intersection type based on traffic parameters with AHP method, **Abdul Majid Mozbanzadeh Dezfouli**
22. analytically evaluation of accident-prone points of the country 's suburban Highway network, a case study: accident-prone points in East Azerbaijan province, **Mostafa Gholami**
23. Development of a surrogate safety index based on the new concept of influence zone for 2D vehicle trajectory, **Mohsen Nakhai, 1397/10/22**
24. Evaluation and comparison of proximity safety measures for vehicles' longitudinal movement, **Fariborz Mansourifard, 1397/11/01**
25. Performance of Base isolations and Dampers in Steel Structures Subjected to Influence of Explosion, **Roham Zakavi, 1398/06/03**
26. Presenting a decision-making model for the management of strategies and solutions to secure suburban roads in Hormozgan province using hierarchical analysis, **Shayan Nadimi, 1397**
27. Evaluation and modeling of damped light steel frame under near filed earthquake, **Roohallah Karimzadeh, 1398/06/03**
28. Safety evaluation based on the traffic conflict technique by the method of before and after observational studies, **Mohammad Moein Jahanshahi Brahoyi, 1398/11/16**
29. Theoretical evaluation of the possibility of rear-end accidents in adaptive cruise control, **Farshad Bashiri, 1398.**
30. Microscopic traffic flow modeling using ANFIS, **Saeed Sekandari, 1399/02/19**
31. Development of a lane-changing driver assistant model based on surrogate safety measures, **Hamid Atot, 1399/02/19**
32. Aggressive Driver Behaviors Modeling Using a Questionnaire, **Behnam Abdi, 1399/11/14**
33. Developing a pseudo-magnet traffic flow model for autonomous vehicles, **Majid Abbaspour, 1399/11/14**
34. Safety assessment of suburban roads with traffic counter data, **Sohail Khademi Bami, 2019.**
35. Designing an adaptive optimal strategy for an intersection using traffic light and vehicle trajectory optimization, **Armin Nikpay, 1399/11/15 (advisor)**
36. Evaluating and ranking effective factors on establishment of intelligent cities from transportation viewpoint - a case study (Shiraz city), **Habib Taghipour, 1399/11/21**
37. Use of fuzzy approach in comparing the objective and subjective safety of rural roads, **Mohamad Askarpour, 1400/11/20 (advisor)**
38. Developing a new surrogate safety measures using natural driving data, **Akram Mazaheri, 1400/11/12 (Doctoral dissertation advisor)**

39. Investigating the safety of rural roads with surrogate safety measures, **Amirhosein Setoudeh**, 1401/06/21
40. Comparison and evaluation of various surrogate safety measures based on real traffic microscopic data, **Mohammad Soltanirad**, 1401/06/21
41. Severity analysis of crashes using structural equation modeling, **Raziyeh Vahdati**, 1401/06/21
42. Modeling the severity of truck crashes using classical econometric and machine learning methods, **Seyed Amir Mohammad Hosseini**, 1402/06/08
43. Analyzing accident severity in different lighting conditions with machine learning method, **Elham Magholi**, 1402/11/18
44. Investigating the relationship between the amount and type of pavement distress and safety of urban roads based on traffic conflict technique, **Mohammad reza Hosseini**, 1402/06/28 (advisor)
45. Factors affecting the choice of active transportation by separating mandatory and optional trips, **Mohammad Aref Yarahamdi**
46. Modeling of ridesharing behavior of travelers using machine learning methods, **Mohammad reza Iranmanesh**
47. Analyzing the use of alternative fuel vehicles with different machine learning methods, **Hamid Khosravi**
48. Development of a web-based tool for the exploratory analysis of accident data, **Amir Khalili Parisi**
49. Using the fuzzy approach in determining the critical threshold of the time-to-collision index (TTC), **Poya Shorafa** (advisor)